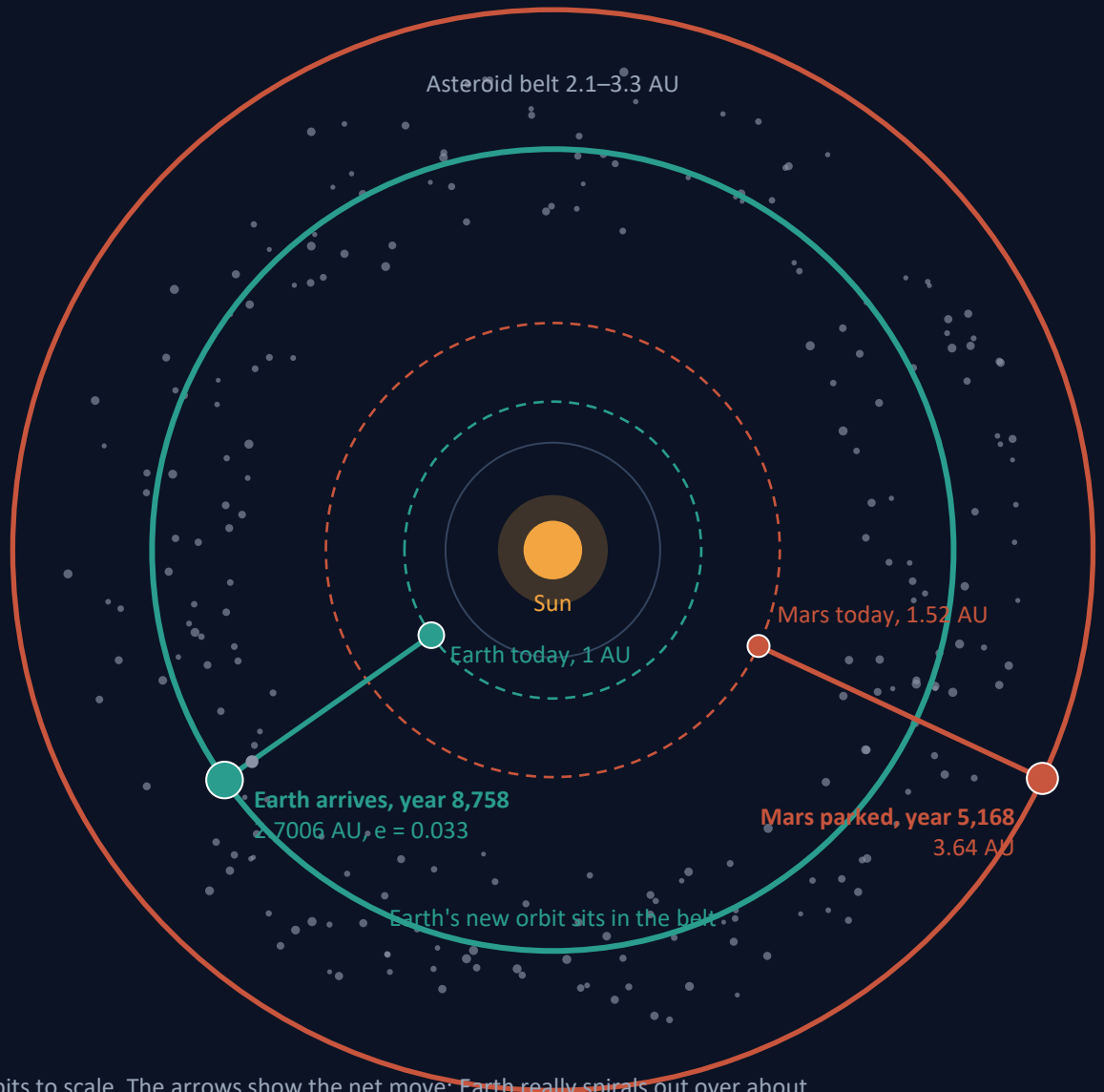
Towing Earth to 2.7 AU

A gravity tractor held by light. An asteroid's gravity pulls Earth outward; focused light on its mirror holds the asteroid in place. Nothing touches the planet.

2.7006 AU Earth's new orbit (target 2.7)	**8,758 yr** arrival, of 10,000 allowed	**10 / 10** official benchmarks pass
1.57 lowest safety factor (needs 1.5)	**5.6×10^{-16}** momentum ledger gap (limit 10^{-9})	**0** unresolved flags (6 covered by E1–E3)

Sushmit Sukla (team leader) · Ashish Bharti · Akhil Semwal · Madhukar Verma · Prabhanshu Kerketa

25 October 2026 · github.com/Ashish000011/black-ant · test bench: ashish000011.github.io/black-ant



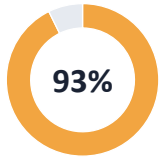
Orbits to scale. The arrows show the net move; Earth really spirals out over about 4,700 laps of the Sun, Mars over 1,700.

All six winning conditions hold

Each condition of the participant guide, measured in the final run (results/summary.json) and checked again from scratch by run.bat verify, which re-runs the benchmarks, the tests, the engine and the studies.

1	**Earth reaches the target orbit within 10,000 years** required: 2.7 AU by year 10,000	**year 8,758** 2.7006 AU · 1,242 years to spare	PASS
2	**Every Earth-safety threshold holds all run** required: all ten, at every 100-day check	**10 of 10** held at all 32,686 checks	PASS
3	**Every structural safety factor ≥ 1.5** sized at each tow's full power cap	**1.57** lowest: cables and anchors	PASS
4	**Every applicable benchmark passes** the guide's ten, inputs changeable	**10 of 10** plus our checks X1–X9 and 49 tests	PASS
5	**Forces booked; energy and momentum kept** ledger within 10^{-9} of the applied impulse	**5.6×10^{-16}** 6.2 orders of magnitude inside	PASS
6	**No unresolved flags** a flag may stand only if an enabler covers it	**0 open** 6 raised, each covered by E1, E2 or E3	PASS

Three declared enablers, each with a hard limit



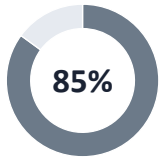
E1 · Power source
9.27×10^{25} W at peak, of 10^{26} W

Covers: beam power above humanity's 2×10^{13} W; Vesta's 0.018 AU pass; belt impact rates.



E2 · Super-mirror material
mirror at 1,466 K of 1,500 K at the power cap

Covers: tow mirrors and cables (absorbs 10^{-7} of the light, 100 GPa strength).



E3 · Planet-scale manufacturing
8.5×10^{8} km² of optics, of 10^{9} km²

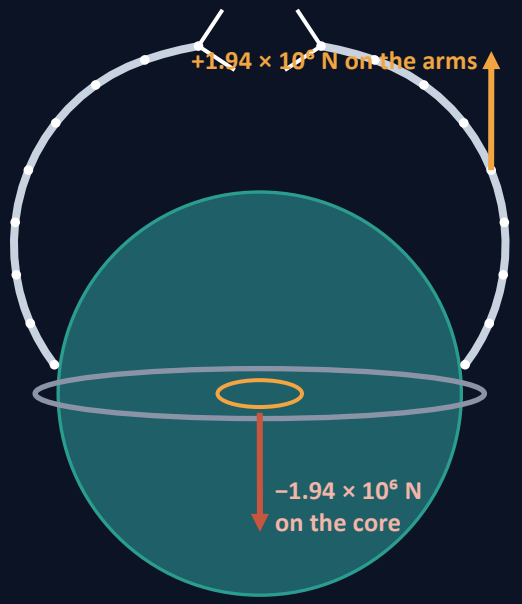
Covers: placing 5.5×10^{20} kg of asteroids; building the mirrors.

Re-checked from scratch: run.bat verify → 10/10 benchmarks · 49 tests · a fresh engine run identical to the official one · the Moon stays bound · 3 studies reproduce (974 numbers, 0 differ) · 6/6 conditions → one results PDF

Why not Octo-Earth: it cannot push, and it cannot stand

We audited the company's eight-armed reference with the same physics as our machine, at the arm its demo draws: 8 arms of 12,000 km in 24 jointed segments, a 200 km tube with 2 km walls carrying 1 MA, curling 72° round the globe.

THE ARMS AS THE DEMO DRAWS THEM (Earth and lengths to scale)



Sum of the two: 3.4×10^{-13} of either force (test X8). Earth's field comes from its own core, so the arms' push on the field comes back on the core: whatever their shape, the arms only push on Earth itself.

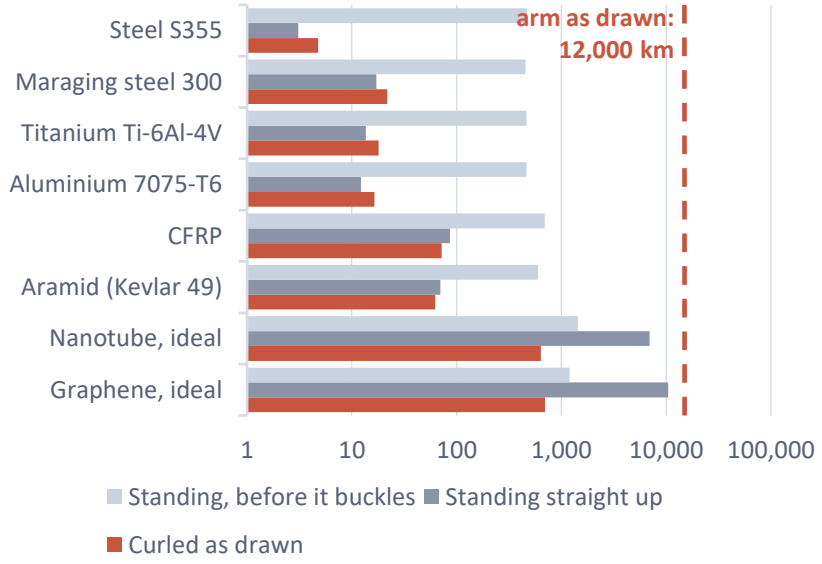
3.4×10^{-13}

net push, as a share of the force
Each side feels 1.94×10^{6} N; they cancel to rounding error. In the demo's 0.5 nT outside field, all closed circuits together feel 2×10^{-12} N.

0.012

safety factor at an arm's root
Curled as drawn, a 12,000 km arm is a beam held only at its root: 5,500 GPa there even in ideal nanotube, whose measured strength is 63 GPa. It needs 1.5.

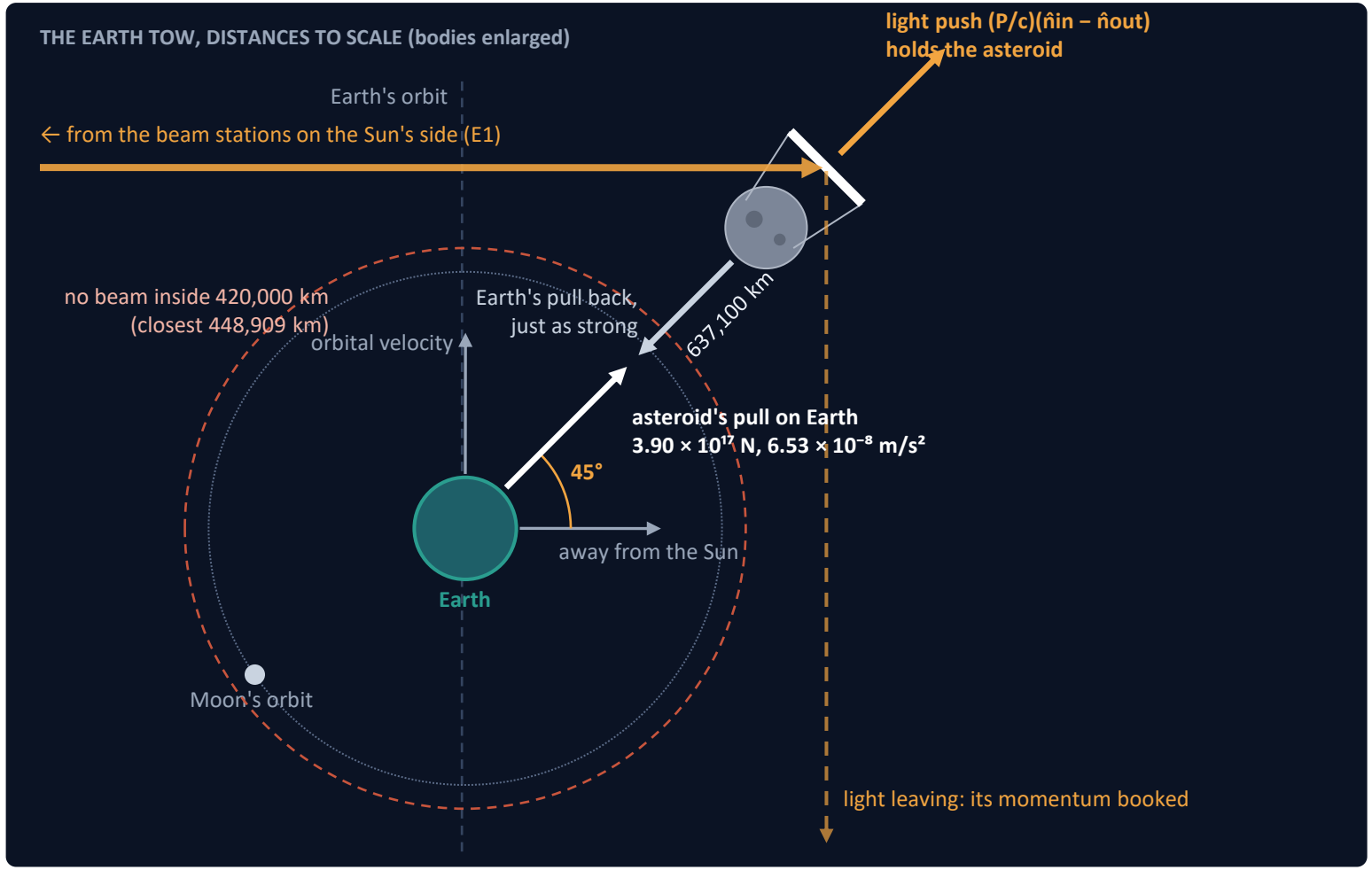
Longest arm that holds itself up, at safety factor 1.5 (km, log scale)



So we move Earth with an outside mass. An asteroid's gravity is a real external force: no structure on the planet, nothing to push against, and every reaction is booked in the ledger.

A gravity tractor held by light

An asteroid 1.66 times the Moon's distance away pulls Earth outward with 3.90×10^{17} N. Earth pulls it back just as hard, so focused light on a mirror tied to the asteroid holds it on its spot. Earth feels only gravity.



1 Tow asteroid
3.97×10^{20} kg, 336 km radius at 2,500 kg/m³; held 637,100–849,900 km from Earth; E3 places it.

2 The pull
3.90×10^{17} N on Earth, 6.53×10^{-8} m/s². Its along-orbit part widens Earth's orbit a little on every lap (Gauss).

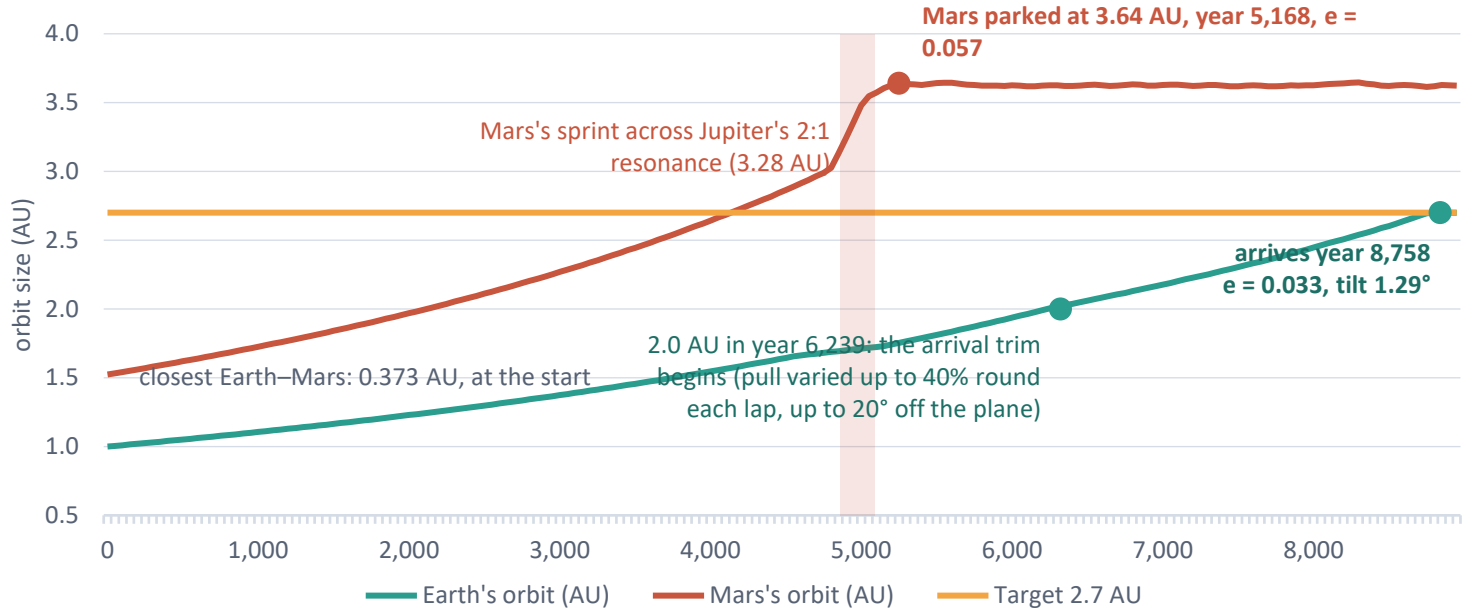
3 The beam (E1)
4.7×10^{25}–8.9×10^{25} W on the Earth tow; both beams peak at 9.27×10^{25} W of 10^{26}; the stations' recoil (up to 3.1×10^{17} N) is booked.

4 The controller
Asks each step for the force that keeps the asteroid on its spot (10-day response); it strayed at most 2,715 km (0.4%).

5 Why 45°
Light can only push. At 45° the pull grows the orbit most per watt; the mirror stayed at 45.1°–48.7°.

1 AU to 2.7006 AU in 8,758 years

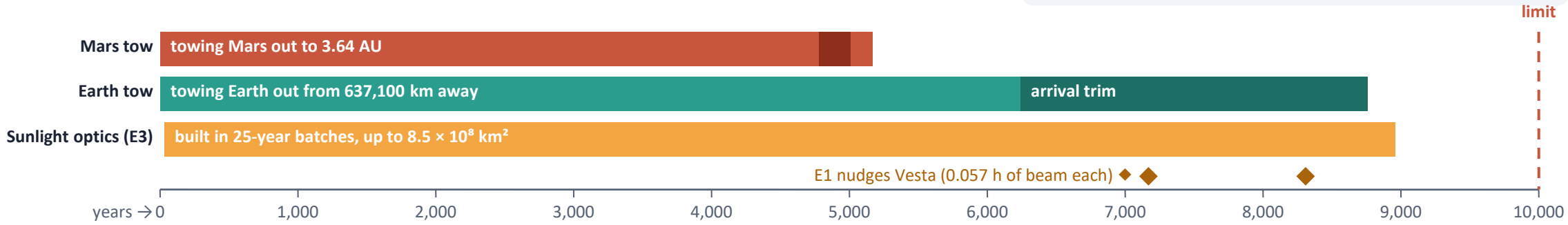
Mars is moved first and parked at 3.64 AU in year 5,168, so the planets never come closer than 0.373 AU. From 2 AU, Earth's tow trims the arrival orbit to the one Jupiter forces at 2.7 AU.



Crossing Jupiter's 2:1 resonance

At 3.28 AU Mars's year is half Jupiter's, and a slow crossing gets caught. Eight starting points per pull; a filled dot never reached its parking orbit.

normal pull	●●●●●●○○	6/8
1.56× sprint	●●○○○○○○	2/8
3.3× sprint	○○○○○○○○	0/8
4.9× sprint	○○○○○○○○	0/8
4× sprint (chosen)	○○○○○○○○	0/8



Every push is booked with its partner

A Newtonian N-body engine with an error-free momentum ledger. The only outside forces are the beams on the two tow asteroids, and their partners (the stations' recoil, the light that leaves) are booked too.

Gravity, every pair once

$$\ddot{r}_i = \Sigma_j\, G\, m_j\, (r_j - r_i)/|r_j - r_i|^3 + F_i/m_i$$

Sun; Venus to Neptune on J2000 orbits; Ceres, Pallas, Vesta, Hygiea; both tow asteroids

Light pressure

$$F = (P/c)(\hat{n}in - \hat{n}out)$$

a mirror is only ever pushed; the power for a wanted force: $P = |F|^2 c / (2\, \hat{n}in \cdot F)$

Orbit growth (Gauss)

$$da/dt = (2a^2 / \mu)\, v \cdot f$$

only the along-orbit part of the pull grows the orbit; benchmark 8 checks it

Climate (Budyko–Sellers)

$$(S/4)(1 - \alpha(T)) = \varepsilon\, \sigma\, T^4$$

albedo 0.30 → 0.62 as ice spreads; a latitude profile gives the liquid-water share

The Moon stays (Hill)

$$r_H = a\, (m\oplus / 3M\odot)\text{^}⅓$$

the Moon at 0.256 of Earth's Hill radius today, 0.095 at 2.7 AU

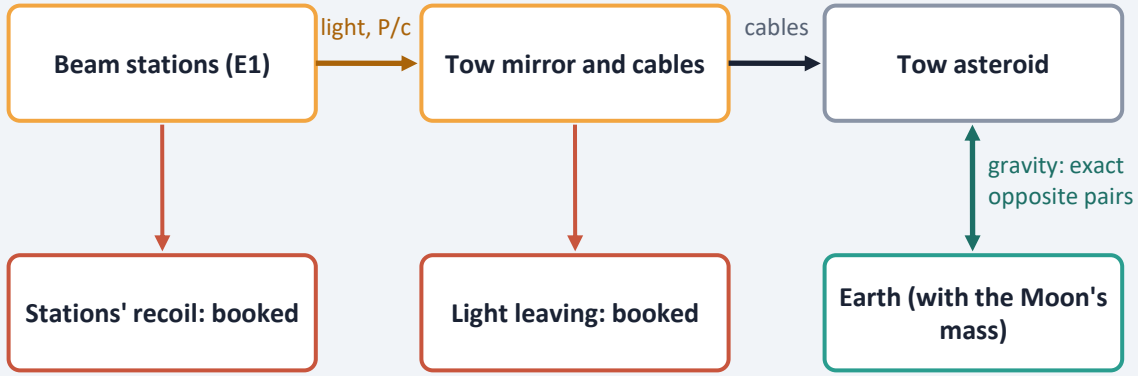
Air and magnetic shield

$$\lambda = GMm / (kT\, r)$$

nitrogen's Jeans λ stays ≥ 194 (escape needs < 30); magnetopause 10 → 13.9 R⊕

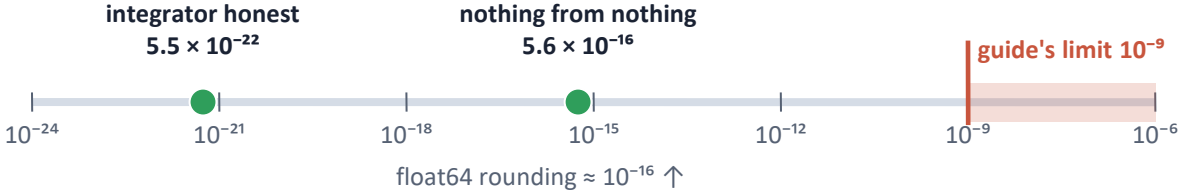
Integrator	Used for
Leapfrog (kick–drift–kick), 1 day	the journey; half the step moves arrival 0.3%
Yoshida 4th order	the ten official benchmarks
REBOUND MERCURIUS, 5 days	a million years after arrival
REBOUND IAS15	an independent cross-check (X6)

WHERE EVERY PUSH GOES



Applied impulse over the journey: 8.78×10^{26} kg m/s. Asteroids sent away after arrival take their momentum with them, booked.

The guide's two ledger checks, worst over the journey (log scale)



Energy: the beams did 1.7969×10^{33} J of work and the bodies gained 1.7970×10^{33} J, counting what left with the asteroids: a 0.007% match.

10 of 10 benchmarks, 49 tests, one command to check it all

Every official benchmark passes, also with inputs a judge changes (python -m sim.benchmarks 4 --set e=0.5); run.bat verify re-runs everything and writes a results PDF.

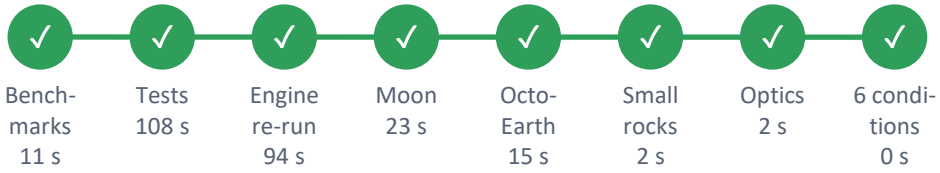
#	Benchmark	Measured	Expected (tolerance)	
1	Wire in a field	(0, 0, 10) N	(0, 0, 10) N (10^{-9})	✓
2	Closed loop, uniform field	4.3×10^{-16} of I L B	0 (10^{-9} of I L B)	✓
3	Long wire field	2.00000×10^{-4} T	$\mu_0 I/2\pi r$ (10^{-6})	✓
4	Kepler orbit back to start	back within 1.8×10^{-8} a; energy 1.1×10^{-10}	10^{-6} a; 10^{-9}	✓
5	Solar system energy, no push	energy error 8.9×10^{-11}	10^{-9}	✓
6	Earth–Moon barycentre period	365.260 days	365.256 (0.1 day)	✓
7	Hohmann transfer	2.946, 2.650 km/s; 258.9 days	2.94, 2.65 km/s; 259 d (±0.02; ±2)	✓
8	Push along velocity, orbit growth	3,169.7 km	3,170.1 km, Gauss (1%)	✓
9	Cantilever under self-weight	root moment, tip deflection within 10^{-9}	$qL^2/2$, $qL^4/8EI$ (1%)	✓
10	Equilibrium temperature	254.0 K	254 K (2 K)	✓

Convergence: halving the step to 0.5 day moves arrival from year 8,758 to 8,730 (0.3%), with the same orbit and 0 unresolved flags.

Our own checks, X1–X9

- **X1** Kepler's third law
- **X2** Energy steady for 1,000 years
- **X3** Ledger with a push on
- **X4** A Hohmann transfer, flown
- **X5** Arrival time under steady push
- **X6** Agrees with REBOUND IAS15
- **X7** The asteroid's pull and hold
- **X8** Octo-Earth: zero net push
- **X9** Earth's real axial-tilt cycle

run.bat verify: the whole check, one command, one PDF



Structures, sized for the most a beam can ever push (2P/c at the power cap), so the catch-up after an outage is covered

Cable safety factor ≥ 1.5
Earth 1.57 · **Mars 1.57**

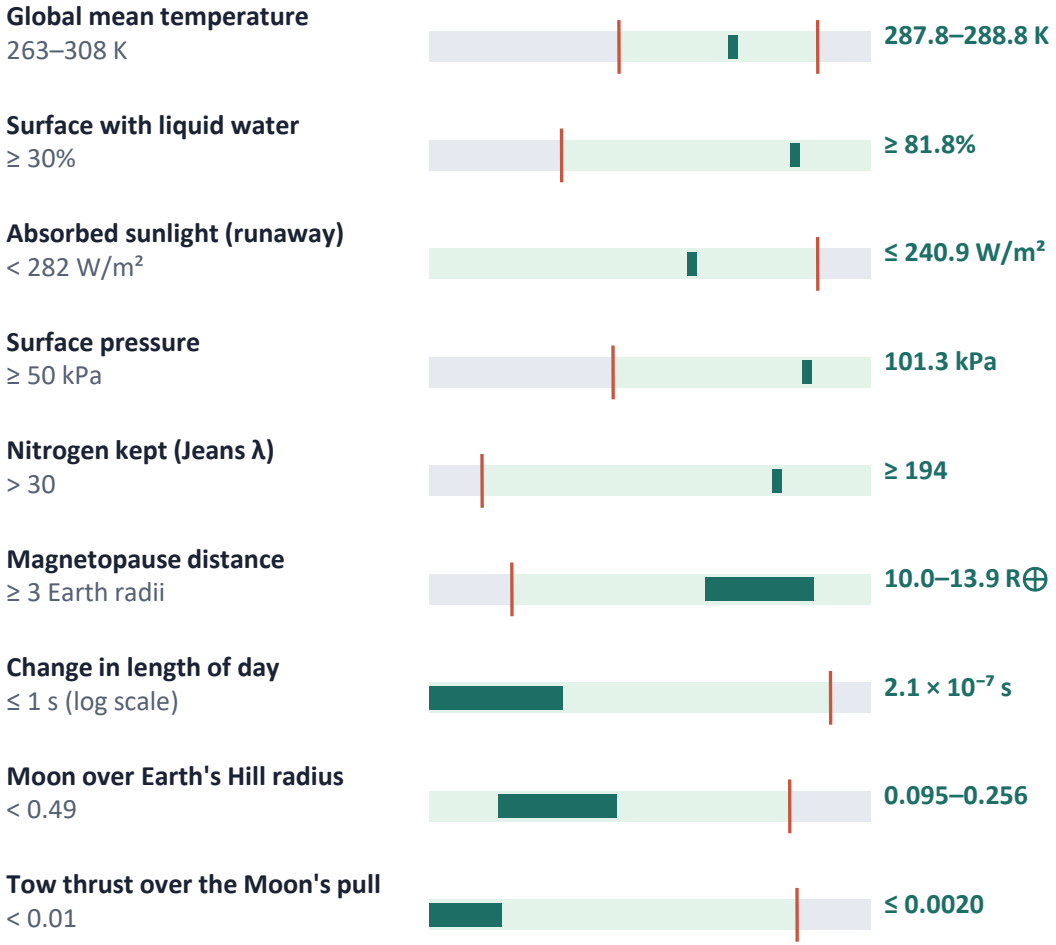
Anchor safety factor ≥ 1.5
Earth 1.57 · **Mars 1.57**

Tow mirror temperature ≤ 1,500 K
Earth 1,466 K · **Mars 1,463 K**

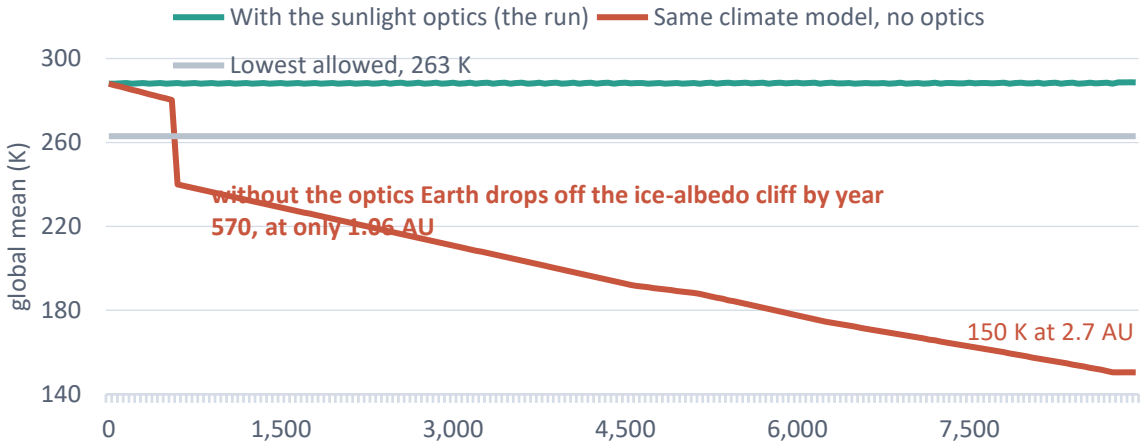
Asteroid body safety factor ≥ 1.5
Earth 120 · **Mars 152**

Ten thresholds held at every one of 32,686 checks

At 2.7 AU Earth gets 13.7% of today's sunlight and the same model freezes it at 150 K. Sunlight optics built by E3 hold it at 0.995–1.010 of today's sunlight, 287.8–288.8 K.

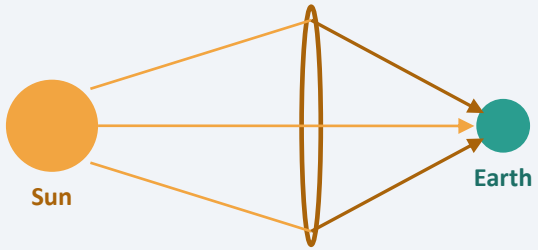


green: allowed · red tick: the limit · teal: the range over the whole journey



WHERE THE OPTICS SIT: A REFRACTING RING NEAR L1

ring near L1, 4.04 M km from Earth



Film bends extra sunlight onto Earth from the Sun's own direction, so noons and nights stay natural (Angel 2006, used the other way round).

Film passes	Sunlight	Earth
95%	97.3%	286.0 K ✓
90%	91.5%	281.7 K ✓
85%	85.9%	238 K, frozen ✗

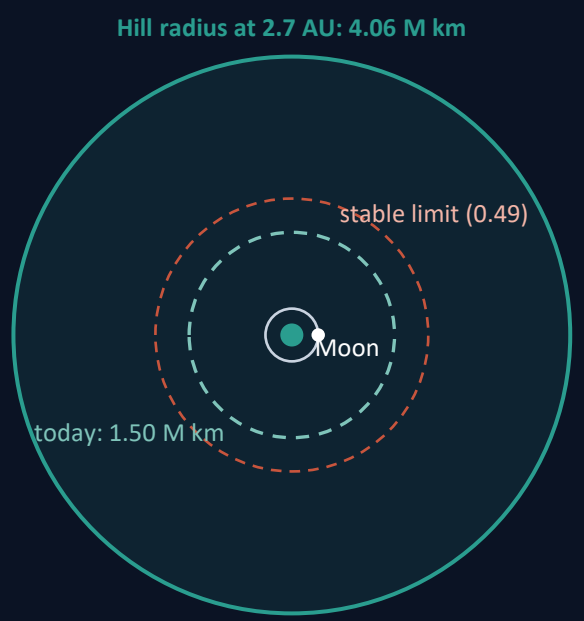
Held to E3's 10^9 km². The climate cliff is at 89.3% of today's sunlight, so the film must pass at least about 88%.

Black Ant · Towing Earth to 2.7 AU · TransfiNITTe 2026 · every number from results/

The Moon stays, and the new orbits hold for a million years

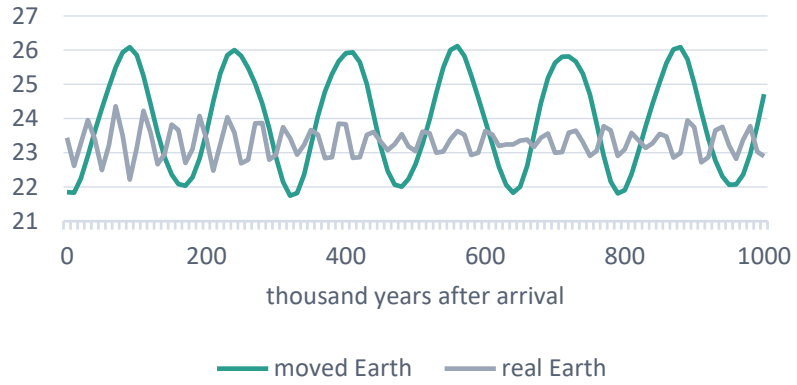
With the tows gone, the moved solar system ran 1,000,000 years beside the real one (REBOUND MERCURIUS): no planet lost, 0 unresolved flags, and 2 chaos copies started with tiny changes agree.

EARTH'S HILL SPHERE, TO SCALE

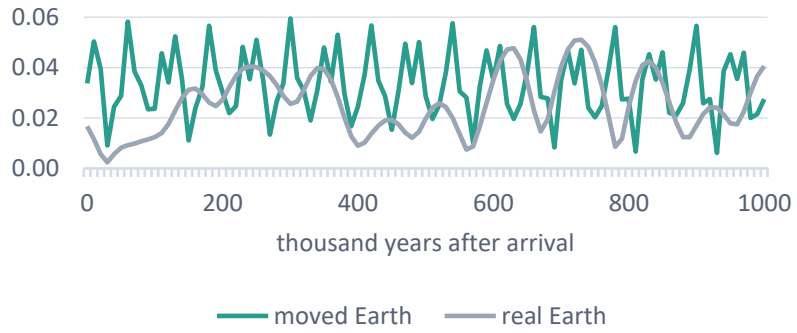


Moon test	Earth at	Asteroid moves the Moon by
start, year 0	1.00 AU	0.049%
middle, year 4,000	1.55 AU	0.023%
end, year 7,999	2.70 AU	0.006%

Axial tilt (degrees)



Eccentricity



Tilt swings 21.7°–26.1° (real 22.2°–24.4°): spin precession nears a slow planetary mode; E3 re-aims 1.4% of the sunlight (a covered flag).

	Moved	Real
Earth's orbit (AU)	2.698–2.702	0.99998–1.00004
Earth's eccentricity, max	0.060	0.051
Earth's tilt	21.7–26.1°	22.2–24.4°
Mars's orbit (AU)	3.614–3.655	1.5236–1.5239
Earth–Mars orbit gap, min	0.36 AU	0.30 AU
Energy error	1.3×10^{-8}	4×10^{-9}

THE SKY AND THE DAYS AT 2.7 AU

Sun 0.20° wide
37% of today's width

Year: 4.44 of ours
each season about 13 months

Same day length
the tow adds 2.1×10^{-7} s

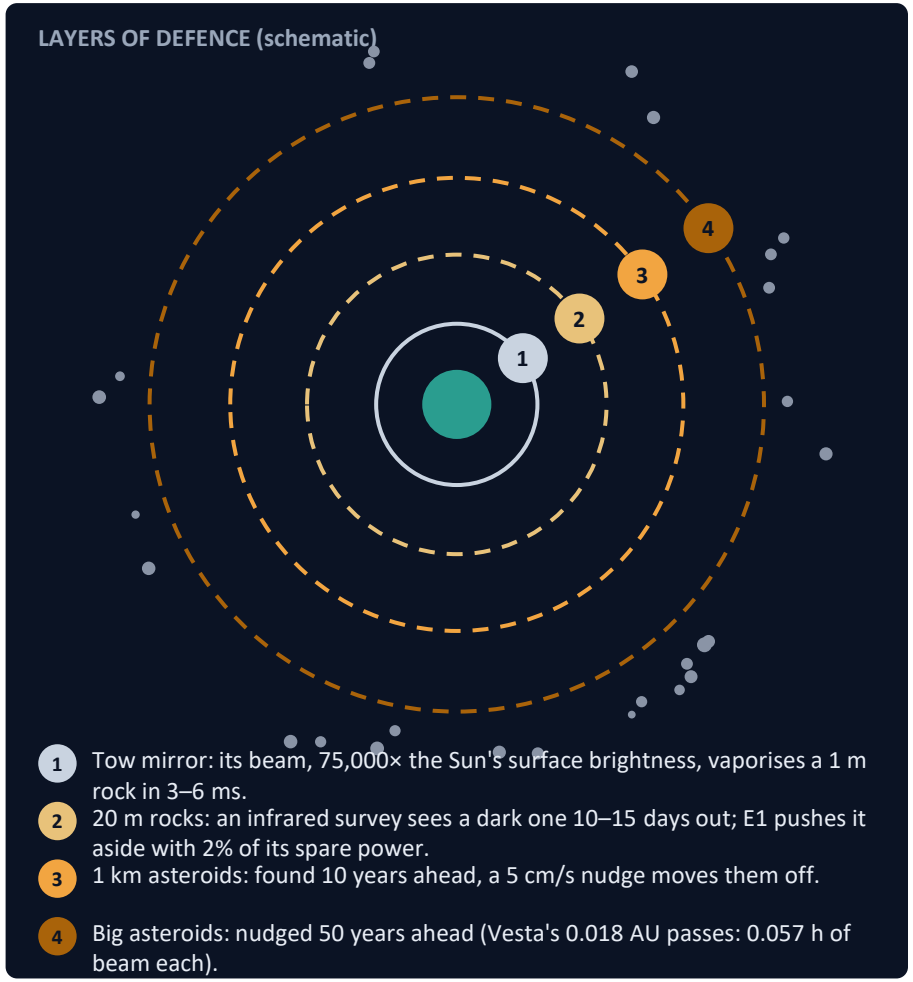
Tow mirror ≈ full Moon
0.47° disc, 135° from the Sun

Stronger shield
magnetopause 10 → 13.9 Earth radii

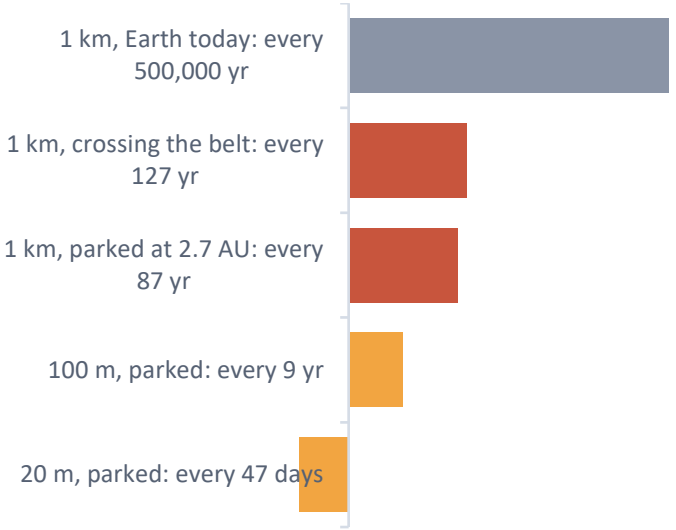
45 big-asteroid passes
each nudged 50 yr ahead (≤ 0.62 h of beam)

Parked in the asteroid belt, and defended

At 2.7 AU Earth sits inside the belt: a 1 km impact every 87 years instead of every 500,000. E1 watches and nudges, the tow mirrors protect themselves, and every beam survives station outages.



Years between impacts (log scale)

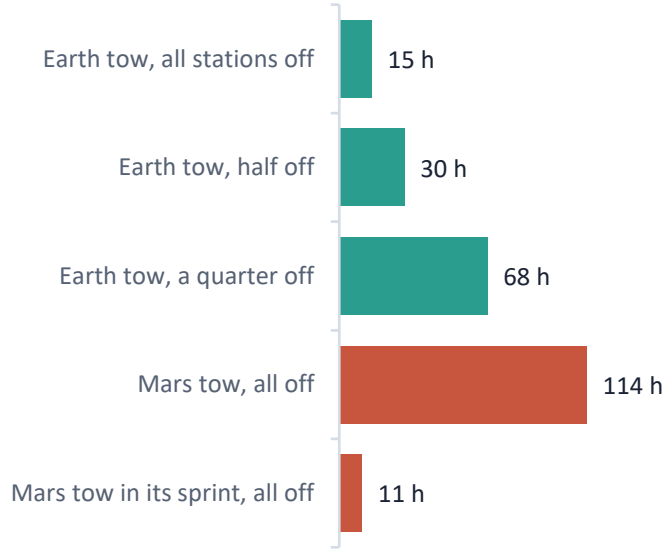


Parked rates: about 8 rocks of 20 m+ a year (range 1.2–40), and a 100 m one every 9 years.

Kirkwood gaps don't help: encounters at 2.50 AU are 0.75×, at 2.82 AU 1.05× those at 2.70: the gaps are in orbit size, not in space.

The Moon, Mars, its moons: get craters only: breaking the Moon up takes 20× the whole belt's mass.

How long each beam can be cut and its asteroid still held



Many independent stations share E1's budget. The structures are sized for the full cap, so catching up after an outage is safe; 7.3% of E1 stays spare for rocks and asteroids, and dust holes about 568 km² of optics a year for E3 to patch.

What the model proves, and what it does not

✓ Proven in simulation

- Earth to 2.7006 AU by year 8,758, with Mars moved first
- ten Earth-safety thresholds held at all 32,686 checks
- every structure at safety factor ≥ 1.57, at full beam power
- 10/10 official benchmarks, checks X1–X9, 49 tests
- momentum ledger closed to 5.6×10^{-16}
- stable for a million years; the Moon stays bound

! Not modelled, or still open

- the hardware: E1–E3 are declared and bounded, not engineered
- climate in one dimension: no weather, oceans or carbon cycle
- 13-month seasons: the model gives yearly means only
- tow-mirror scatter must stay below 3.4×10^{-11} of the beam; no enabler bounds it
- the Moon and Earth's spin run separately from the journey
- the safety thresholds are the company demo's; a new rulebook means a re-run

→ Next

- the Moon as its own body through the whole journey
- a climate resolved by latitude and season
- a sensitivity sweep of tow mass and distance
- a scatter specification and test for the tow mirrors
- resonance transport in the belt with 20,000+ orbits
- aimable mirrors re-aiming 1.4% of the sunlight for the seasons
- re-run with the organisers' thresholds the day they publish

Check us yourself. `run.bat verify` re-runs the benchmarks, the tests, the engine and the studies and writes a results PDF in about 4 minutes; `python -m sim.benchmarks 4 --set e=0.5` changes a benchmark's inputs. Test bench: ashish000011.github.io/black-ant · Code: github.com/Ashish000011/black-ant

Key sources: Lu & Love 2005 (Nature) · Korycansky, Laughlin & Adams 2001 · Angel 2006 (PNAS) · Budyko 1969 · Sellers 1969 · Goldblatt et al. 2013 · Rein & Liu 2012 (REBOUND) · Yoshida 1990 · Burdanov et al. 2025 · Lubin et al. 2014 (DE-STAR); 43 in the technical report.

Team Black Ant · Sushmit Sukla (team leader) · Ashish Bharti · Akhil Semwal · Madhukar Verma · Prabhanshu Kerketa